

ACS101 — Programming Languages

Tutorial Hour: Week-by-Week Practice Problems

Week	Language	Tutorial Problems
1	C	1. Print your name and branch. 2. Add two numbers entered by user. 3. Find area of a rectangle.
2	C	1. Swap two numbers. 2. Check if a number is even or odd. 3. Simple calculator (+, −, ×, ÷) using if-else.
3	C	1. Find the largest of three numbers. 2. Grade calculator (marks → A/B/C/D/F). 3. Day name using switch-case.
4	C	1. Print multiplication table of N. 2. Sum of first N natural numbers using loop. 3. Count digits in a number.
5	C	1. Print prime numbers from 1 to 50. 2. Fibonacci series (first 10 terms). 3. Reverse a number.
6	C	1. Factorial using function. 2. Check palindrome number using function. 3. Find GCD of two numbers.
7	C	1. Recursive factorial. 2. Recursive Fibonacci. 3. Sum of digits using recursion.
8	C	1. Input array of N numbers; find max and min. 2. Compute average of array. 3. Linear search in array.
9	C	1. Count vowels in a string. 2. Reverse a string. 3. Check if string is palindrome.
10	C	1. Store 5 numbers in an array and print them in reverse order. 2. Basic string operations — length, copying, concatenation.
11	Python	1. Hello World in Python. 2. BMI calculator. 3. Temperature converter ($^{\circ}\text{C} \leftrightarrow ^{\circ}\text{F}$). Compare C vs Python syntax.
12	Python	1. Grade calculator in Python. 2. Fibonacci using Python loop. 3. Simple calculator using functions in Python.
13	Python	1. List: marks of 5 students — max, min, average. 2. Dictionary: store name and marks of 3 students. 3. Count word frequency in a sentence.
14	Python	1. Write student records to a text file. 2. Read and display the file. 3. Read a small CSV and print rows.
15	Python	1. NumPy: create array, compute mean. 2. Matplotlib: plot a line graph of marks. 3. Mini-demo: simulate 5 IoT temperature readings and plot.